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How size-dependent settling can both bias and inform microplastics observations

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Abstract

Particle settling and rise velocities play a central role in modeling microplastic (MP) transport, residence times, and distributions in aquatic systems. Recent studies have focused on deriving accurate models for MP settling velocities, but the detailed particle-shape information required is usually unavailable in real-world data. Here, we investigate the relative importance of settling velocity model choice versus particle size and shape information using a previously published laboratory dataset of measured settling velocities for irregular MP fragments. We find that uncertainty in particle geometry dominates model error, while differences between commonly used settling velocity models are comparatively small. Settling and rise velocities vary widely across realistic MP sizes and shapes, which leads to orders-of-magnitude differences in residence times. We show how this differential settling can alter observed particle size distributions: for example, even when the underlying fragmentation spectrum follows a power law, we find separate power-law exponents for large and small particles due to differential vertical transport. Our results imply that improving geometric characterization of particles is more critical for modeling vertical transport than further refinement of settling formulas, and that commonly used power-law extrapolations across size classes may be physically inconsistent. These findings have direct implications for interpreting MP observations and for the harmonization and risk-assessment frameworks that rely on size-distribution corrections.

1. Introduction

Concern about microplastics (MPs) in the environment has grown alongside our ability to measure and quantify MP pollution. However, obtaining MP observations remains challenging and labor intensive, resulting in sparse measurements over space and time. Thus, modeling has become a necessary tool to interpret and extrapolate these limited observations. For example, MP ocean inputs have been modeled from waste generation, infrastructure, and population (Jambeck *et al* 2015, Lebreton *et al* 2017, Borrelle *et al* 2020, Weiss *et al* 2021), and ocean and lake surface and water column distributions of plastic have been estimated using hydrodynamic models (Kukulka *et al* 2012, Lebreton *et al* 2019, Daily and Hoffman 2020, Kaandorp *et al* 2020, 2023, Onink *et al* 2021). More recently, modeling is being used to extrapolate the size distribution and polymer diversity of MPs from samples in order to quantify exposure and risk to humans and other organisms (Koelmans *et al* 2020, Mehinto *et al* 2022, Hataley *et al* 2023). Modeling is necessary to inform policy and management, and yet, these models are built on a variety of assumptions each associated with their own uncertainties, and it is not always clear how these uncertainties compound through the modeling choices. This leads to questions about the relative importance of pure model error, uncertain particle information, and measurement errors on model accuracy and uncertainty.

We aim to address some of these questions by focusing on a specific process that is frequently modeled and serves as a sub-model for more complex simulations of MP transport: the settling and rise velocities of MPs. Among the many important transport processes, settling plays a crucial role in how long MPs stay in the water column, which in turn dictates bio-availability and transport (Daily *et al* 2022, Kaandorp *et al* 2023). As a result, considerable emphasis has been placed on accurate predictions of MP settling and rise velocities because of their importance in the fate and transport of MPs in the environment (Waldschläger and Schüttrumpf 2019, Yu *et al* 2022, Coyle *et al* 2023, Dittmar *et al* 2024). They are also an interesting case because, in practice, these models either rely on accurate geometric descriptions of the MPs that are not widely available in environmental data (Waldschläger and Schüttrumpf 2019, Yu *et al* 2022) or use very idealized descriptions (Van Sebille *et al* 2015, Lebreton *et al* 2019). Much has been written about the diversity of MPs, but the majority of the focus has been on diversity of polymer, additives, and morphology (Rochman *et al* 2019). Size and shape play a potential major role in MP transport as well as ingestion potential (Bagheri and Bonadonna 2016, Mehinto 2022). Because toxicity scales with ingested particle number and surface area, smaller MPs are a greater concern, yet most measurements are of larger MPs. This leads to a fundamental challenge in MP risk assessment: small MPs are hard to measure but are likely more relevant to toxicity.

There are efforts to extrapolate the number of small particles from measurements of large particles (Koelmans *et al* 2020, Mehinto 2022), but these approaches can implicitly assume that the transport of all MPs is similar such that MPs will continue to move together as they fragment into smaller and smaller size classes. This is not necessarily true, as differential transport is expected depending on the characteristics of the MPs (Laxague *et al* 2018, Kaandorp *et al* 2021). For example, as MPs become smaller, they become much more 'tracer-like'. In other words, their settling/rise velocities become small relative to the fluid flow and they effectively trace the fluid velocity. Conversely, larger MPs are more 'non-tracer-like': their settling/rise velocities are large relative to the fluid flow, such that their transport is dominated by gravity rather than by the fluid motion. The degree to which a certain MP may behave as a tracer inherently depends on its fluid environment (Chamecki *et al* 2019, DiBenedetto 2026). This disconnect between a tracer and a settling/rising regime of MPs drives size-selective sorting that can arise during transport. A few studies have examined the combination of fragmentation and transport on a large scale, finding that both local fragmentation and sorting via transport play an important role in the particle size distributions (PSDs) present in different environmental compartments (Isobe *et al* 2014, Kaandorp *et al* 2021).

Many studies have investigated the settling of MP particles in laboratory settings (Kaiser *et al* 2019, Waldschläger and Schüttrumpf 2019, Van Melkebeke *et al* 2020, Francalanci *et al* 2021, Dittmar *et al* 2024), though most of these studies have used larger MP particles that are 1 mm to several milimeters in length, that are usually lab generated. Several studies have compared the performance of MP-specific models with more general settling models derived for natural particles (Van Melkebeke *et al* 2020, Coyle *et al* 2023, Dittmar *et al* 2024). In general, these studies have found that settling equations designed for other particles, such as volcanic ash (Bagheri and Bonadonna 2016), perform as well, if not better, than plastic-specific equations. Recent work has shown that many of these models perform worse when applied to small MPs, which makes sense given the difference in Reynolds numbers as compared to the large MPs they were trained on (Dittmar *et al* 2024). (Physically, this is because the drag force on a particle is linear at low Reynolds number and nonlinear at intermediate and large Reynolds numbers (Clift *et al* 2005).) While many of these studies cite the importance of accurate settling equations for use in transport models of the ocean and other large bodies of water (Coyle *et al* 2023, Dittmar *et al* 2024), most large-scale modeling studies have used a modification of the Dietrich (1982) equation that was presented in the work of Kooi *et al* (2017) in modeling settling of plastic in oceans and large lakes (Daily and Hoffman 2020, Lobelle *et al* 2021, Daily *et al* 2022, Fischer *et al* 2022, Kaandorp *et al* 2023).

One challenge in using different settling equations for modeling plastic is that the equations each rely on different size and shape parameters. Most equations require three-dimensional information about the particle, such as an estimate of the equivalent spherical diameter (the diameter of the sphere with the same volume as the particle) or the surface area (Bagheri *et al* 2015, Coyle *et al* 2023, Dittmar *et al* 2024). This 3D information is in addition to the 1D measurements of the longest, intermediate, and shortest diameters of the MP particle. As a result, the data used to derive settling equations are collected using sophisticated camera setups to capture information from one or multiple viewpoints (Van Melkebeke *et al* 2020, Dittmar *et al* 2024) but can also include more advanced imaging techniques like SEM and micro-CT (Bagheri *et al* 2015).

Given only the three diameters, one can assume that the particle can be approximated by an ellipsoid and then the volume, surface area, and any other descriptors can be derived accordingly (Van Melkebeke *et al* 2020, Dittmar *et al* 2024). Estimating these 3D parameters with 1D parameters, however, has been shown to be less accurate than using 2D (area) information from cameras (Bagheri *et al* 2015). Unfortunately, 3D or 2D information is almost never published as part of MP sampling. Most sampling studies of MPs publish the longest length of the particle and, at most, another piece of information such as the length of an additional axis (not guaranteed to be either the intermediate or shortest diameter). The shape of the MPs is important not only in calculating settling velocity, but it is also a crucial part of risk assessment (Koelmans *et al* 2020, Mehinto 2022). This has led to the derivation of probability density functions for environmental MP length and other descriptors like volume, surface area, and length-/width ratio (Kooi *et al* 2021).

The strong size-dependence of MP settling and rise velocities implies that particle size directly controls vertical transport, and therefore that particles of different sizes are not expected to behave identically in the environment. This creates a challenge for both measuring and interpreting MP PSDs. It is commonly assumed that fragmentation generates a power-law PSD (e.g. Koelmans *et al* (2020), Kooi *et al* (2021), Mehinto (2022)), but even if such a distribution exists locally in a closed system, it does not necessarily follow that the same power law will be observed in discrete environmental samples. Size-dependent differential transport can decouple particles of different sizes, leading to observed PSDs that differ systematically from the underlying fragmentation spectrum.

This issue is particularly relevant because many current interpretation and risk assessment frameworks implicitly assume that MPs of different sizes are retained together within an environmental compartment, such that the local PSD is controlled only by fragmentation processes. In practice, this assumption underlies commonly used correction approaches that harmonize samples collected with different methodologies by extrapolating from measured size ranges to a consistent size range (typically 1–5000 μm) using a power-law PSD (Kooi *et al* 2021). Such corrections can be extremely large: for example, concentrations derived from manta trawl samples may be multiplied by factors of order 10^4 (Hataley *et al* 2023). However, if vertical transport is size dependent, then particles of different sizes may not co-occur within the same sampling volume, and the assumption of a single power-law PSD within a given environmental compartment may be physically inconsistent. Previous work using box models has demonstrated that fragmentation and transport jointly shape observed PSDs across environmental settings such as coastal and open-ocean regions (Kaandorp *et al* 2021).

While recent studies have focused on refining settling velocity models specifically for MPs, the relative impact of model selection versus uncertainty in particle information is not well-quantified. In addition, we consider the potential bias introduced to MP observations from size-dependent settling. Specifically, in this paper, we address two distinct but related questions: how important is the choice of settling velocity model given realistic information about MP geometry, and how does this size-dependent settling velocity impact observed PSDs. Both questions address the same practical limitation, that the physical behavior of MPs in the water column is controlled by particle geometry and density in ways that are hard to capture from traditional environmental samples. We first quantify how uncertainty in particle size and shape propagates into predictions of MP settling and rise velocities. We find that these predictions are much more sensitive to measurements of size and shape than to the choice of settling velocity model. We then examine how size- and shape-dependent settling leads to large differences in particle residence times, highlighting the large discrepancy in residence time between the larger particles commonly measured and the smaller particles relevant to organism health risk. Finally, we develop an analytical and numerical framework to demonstrate how size-dependent vertical transport modifies observed PSDs, even when the underlying fragmentation spectrum follows a power law. Following these results, we discuss their implications in the context of environmental sampling and risk assessment frameworks.

2. Methodology

We have two main approaches to investigate the relative importance of uncertain particle information in settling velocity calculations and the implications for PSDs. The first is gradually reducing the amount of information available for particle measurements and investigating the errors in the resulting computed settling velocity from three equations. The second is exploring what happens if there is a power-law distribution for idealized particles but they are sampled in only a small vertical range (e.g. with a manta trawl). We describe both approaches below.

2.1. Theoretical framework

First, we briefly review the relevant physics governing particle settling to highlight the importance of MP geometry. The terminal settling/rise velocity of a particle is the particle's vertical speed when the drag force and the net gravitational force are in balance. Here, the net gravitational force also includes the effects of buoyancy. We express this balance of forces as:

$$\frac{1}{2}\rho_f C_D A w_s^2 = (\rho_f - \rho_p)\, g \forall, \quad (1)$$

where ρ_f is the fluid density, ρ_p is the particle density, C_D is the particle drag coefficient, A is the particle's projected area, w_s is the particle terminal settling velocity, g is acceleration due to gravity, and $\forall$ is the particle's volume. If we express particle volume in terms of equivalent spherical diameter d_{eq} where $\forall = \pi d_{eq}^3/6$ and solve for w_s, we arrive at the following general expression for particle terminal velocity:

$$w_s = \sqrt{\frac{\pi g d_{eq}^3}{3 C_D A}\left(\frac{\rho_p}{\rho_f} - 1\right)}. \quad (2)$$

From this expression, we directly see the importance of particle information in predicting velocity: we need accurate particle geometry to give d_{eq} and A, we need the particle's specific gravity ρ_p/ρ_f, and we need the particle's drag coefficient C_D. The drag coefficient is a function of the particle's shape, roughness, and the Reynolds number given by $Re = d_{eq} w_s/\nu$ where ν is the fluid kinematic viscosity. The drag coefficient can additionally be a function of the orientation of a non-spherical particle, but in this study we only consider the average drag coefficient. While C_D can be derived theoretically for ideal geometries at low Reynolds number, most applications use values derived empirically from experiments. Because C_D is a function of Re and thus w_s, solving for w_s can involve an implicit solver. Note that this expression applies to any particle settling velocity and is not specific to MPs. Therefore, we do not necessarily expect 'MP-specific' settling velocity models to perform better than generic models.

2.2. Reconciling measurement errors and settling equations

2.2.1. Particle data

We are using data that was discussed in the study of Dittmar *et al* (2024) and published in a Zenodo repository (https://zenodo.org/records/16021332) that includes data from 3603 MP fragments of PS, PMMA, PVC, and PET (Dittmar 2023). The particles were created using various milling techniques and the fibers were created using the protocol of Cole (2016) with full details provided in Dittmar *et al* (2024). In this study we only look at the fragments and use the QC restrictions suggested by Dittmar *et al* (2024), which results in 3365 fragments. The extracted average settling velocity from the particle tracks is included, along with the longest, intermediate, and shortest particle dimensions, and estimates of the particle sphericity and equivalent spherical diameter (Dittmar *et al* 2024). These variables are estimated from contours of the particles obtained from camera images of the settling track (Dittmar *et al* 2024). We performed two sets of idealized experiments as described below.

2.2.2. Importance of particle size/shape vs settling equation

To investigate the importance of accurate geometric information versus accurate equations for settling velocity, we calculated the difference between predicted and measured settling velocity for the fragments using three different settling velocity equations and five different assumptions about particle shape.

2.2.3. Settling equations

For settling equations, we use the equations of Bagheri and Bonadonna (2016), Yu *et al* (2022), and Kooi *et al* (2017). These three equations are selected because they represent a range of complexity, use, and input. The Bagheri and Bonadonna (2016) equation was derived based on volcanic ash particles and has been shown to perform as well as, or better than, other models for plastic particles (Coyle *et al* 2023, Dittmar *et al* 2024). The model of Yu *et al* (2022), was derived specifically for MPs and has also performed well in comparisons (Coyle *et al* 2023, Dittmar *et al* 2024). The equation of Kooi *et al* (2017) has been used in many large-scale ocean and lake models that consider settling or biofouling (Kooi *et al* 2017, Daily and Hoffman 2020, Lobelle *et al* 2021, Fischer *et al* 2022). Detailed information about the equations can be found in the Supplemental Information.

2.2.4. Data withholding experiments

We use five different assumptions about the MP shape where progressively less information about the true particle shape is known.

The first, called 'All Info,' includes all the information in the dataset from Dittmar *et al* (2024). This includes measurements of particle length, width, and height, as well as an estimate of equivalent spherical diameter and sphericity based on 2D particle images. Thus, we expect this case to provide the best

Table 1. A summary of the five data withholding experiments that were run in this study.

Experiment name	Description	Major assumption
All Info	All of the information of particle shape in Dittmar *et al* (2024) as computed from the image analysis and provided in their dataset.	None
Ellipsoid	The particle is ellipsoidal and we have measurements of all three minor axes (a, b, and c)	$d_{\mathrm{eq}} = \sqrt[3]{abc}$
$b = c$	The particle is ellipsoidal and we only know two axes so we assume the two smallest axes are the same	$b = c$
$a/b = b/c$	The particle is ellipsoidal and we only know two axes so we assume the ratio of those two axes is the same as the ratio of the two smallest axes	$\frac{a}{b} = \frac{b}{c}$
Sphere	We only know one axis measurement and we assume the particle is a sphere	$a = b = c$

estimate of the measured settling velocities. For the other assumptions, we replace measured variables with assumptions commonly used when those measurements are not available.

First, we replace the Dittmar values for equivalent spherical diameter and sphericity with the derived values assuming the particle is an ellipsoid with semiaxes a, b, and c. This computation for d_{eq} ($d_{\mathrm{eq}} = \sqrt[3]{abc}$) was used in Dittmar *et al* (2024) and they found that it decreased the mean relative absolute error by 4%–30% depending on the equation used. Here, sphericity is calculated as the ratio of the surface area of the sphere with diameter d_{eq} to the surface area of the ellipsoid with semiaxes a, b, and c.

Next, we assume that we do not have the height, c, of the particle. This is the most common situation when dealing with environmental samples of MPs, because MPs are usually identified using microscopy methods that give a 2D image. One of two assumptions is usually made. The simplest assumption is that the width and height are the same (called 'b = c' here), which has been used in the literature mainly for regular shapes such as prisms, cylinders, or fibers (Khatmullina and Isachenko 2017). While we expect this to be less accurate for irregular fragments, we include it for its simplicity and to see how sensitive the results are to different choices. The other assumption, which is used in MP risk assessment (Koelmans *et al* 2020, Mehinto 2022), is that the length-width ratio is the same as the width-height ratio (called '$a/b = b/c$'). Finally, we assume that we only have the particle length, a, and the particle is a sphere (called 'Sphere') with diameter a. A summary of the data-withholding experiments is presented in table 1.

We compute the error in the predicted settling velocity for each particle by comparing the value generated by the equation with the measured settling velocity from Dittmar *et al* (2024). We use average absolute relative error percentage as the metric for comparing the accuracy of different estimates. The average absolute relative error percentage is computed as

$$|\mathrm{AE}| = 100 \times \frac{\sum_{i=1}^{N} \left| \frac{w_{\mathrm{meas},i} - w_{\mathrm{pred},i}}{w_{\mathrm{meas},i}} \right|}{N}$$

and is the same metric as used in Dittmar *et al* (2024).

2.2.5. Variation of settling velocity with particle size

Many large-scale modeling studies do not consider size differences in the transport of particles (Van Sebille *et al* 2012, Daily and Hoffman 2020, Morales-Caselles *et al* 2021, Onink *et al* 2021), though a few have indicated that size differences impact the transport of particles (Kooi *et al* 2017, Kaandorp *et al* 2020, Fischer *et al* 2022). Most laboratory-based settling studies use only large particles, so existing studies have not measured the settling velocity of small particles. To visualize the potential differences in particle settling velocity induced by size and shape differences, we take each of the 3365 fragments from Dittmar *et al* (2024) that were used above and rescale all three semiaxes of the particle so that the major axis takes every integer value between 1 and 5000 μm. That is, for each of the 3365 real particles we generate 5000 new particles by assuming that the ratios of the axes remain fixed as the largest axis is scaled to a given size. This results in 3365 theoretical particles for each size between 1 and 5000 μm. We acknowledge that, in reality, large particles likely have different aspect ratios than small particles, but we argue that this is a reasonable assumption for our theoretical exercise. Furthermore, since detailed shape measurements of small particles in environmental settings are not currently available, we cannot compare our shape distribution to one found in real samples. However, the usual assumption is that fragmentation processes will lead to smaller particles being more spherical. In this case, our experiment would be an overestimate of the sensitivity of settling velocity to shape for small particles. For each of these particles, we compute the settling velocity by assuming the particle is an ellipsoid and using the equation of Yu *et al* (2022) because it is an explicit equation that performed well in our testing.

2.3. Particle-size effects on observed PSDs

To investigate the potential impact of size-dependent settling and rise velocities on the observed PSDs, we begin by assuming that fragmentation produces an ideal power-law distribution and then compute how that distribution is modified by vertical transport processes. We consider both buoyant and non-buoyant particles. For simplicity, we assume that particles are defined by their diameter d. We define a generic power-law PSD:

$$p_0(d) = Cd^{-\alpha}, \tag{3}$$

where C is a constant. We assume spherical particles for this analysis, but later apply the methods to nonspherical particles.

We ask how the *observed* PSD is modified by the size-dependent vertical transport for settling and buoyant particles. Assuming horizontal homogeneity, we consider a general distribution where $p(d, z, t)$ denotes the concentration (or number density) of particles of size d at depth z (with $z = 0$ at the free surface and $z < 0$ within the water column) at time t. Given a size-dependent terminal velocity $w_s(d)$ and turbulent diffusion with constant eddy diffusivity K, the vertical evolution obeys an advection–diffusion equation:

$$\frac{\partial p}{\partial t} + \frac{\partial}{\partial z}(w_s p) = \frac{\partial}{\partial z}\left(K\frac{\partial p}{\partial z}\right). \tag{4}$$

Although we consider a constant diffusivity value here for simplicity, this analysis can be readily extended to a generic diffusivity depth profile.

For buoyant particles ($w_s > 0$), we analyze particles near the surface under steady state ($\partial p/\partial t = 0$). For our boundary conditions, we impose no-flux at the surface given by $w_s p - K\partial p/\partial z = 0$ at $z = 0$. For our lower bound, we set $p \to 0$ as $z \to -\infty$. This leaves us with:

$$K\frac{\partial p}{\partial z} = w_s p, \tag{5}$$

where $p = p(d, z)$ and $w_s = w_s(d)$. Solving this equation gives an exponential vertical profile for each size class,

$$p(d, z) = p_0(d)\, g(z|d), \qquad g(z|d) = \frac{w_s(d)}{K}\exp\left(\frac{w_s(d)}{K}z\right), \quad z \leqslant 0, \tag{6}$$

where all of the depth dependence is given by the size-dependent function $g(z|d)$. This function is normalized such that $\int_{-\infty}^{0} g(z|\mathrm{d})\,\mathrm{d}z = 1$, and therefore we can recover the total PSD via depth integration, $\int_{-\infty}^{0} p(\mathrm{d}, z)\,\mathrm{d}z = p_0(d)$. Note that equation (6) is a general form of the model derived in Kukulka *et al* (2012).

To simulate settling particles over time, instead of solving the equation analytically, we numerically simulate particle trajectories. We use a stochastic time-stepping scheme for particle vertical position over time $z(t)$:

$$z(t + \Delta t) = z(t) + w_s\Delta t + R\sqrt{2K\Delta t} \tag{7}$$

where R is a random number with zero mean and unit variance. This applies to particles with constant settling velocity w_s in a fluid with constant diffusivity K.

3. Results

All three settling equations that we tested performed similarly well across all testing (figure 1). This was somewhat surprising, as Dittmar *et al* (2024) described differences in performance when comparing a larger number of settling velocity equations. There are some differences when more information is present. The equations from Bagheri and Bonadonna (2016) and Yu *et al* (2022) perform better than the equation from Kooi *et al* (2017) in the 'All Info,' 'Ellipsoid,' and 'a/b = b/c' cases (figure 1). This is not surprising, because the Bagheri and Bonadonna (2016) and Yu *et al* (2022) equations are tuned to take into account more detailed shape information. When less information is available, as in the 'b = c' and 'Sphere' cases, all three equations perform similarly and the equation from Kooi *et al* (2017) slightly outperforms one or both of the other equations.

As information is removed, the error increases as the computed settling velocities overestimate the measured settling velocities by a larger margin. All equations show similar degradation in performance as information is removed. When only two dimensions of the particle are known, the methods all have an error between 151.5% and 160.8% and this jumps to 371.5%-386.8% when only the length is known (figure 1). This makes it clear that the choice of equation used to calculate the settling velocity of MP particles is less important than the amount of shape information available.

3.1. Importance of particle size/shape vs settling equation

3.1.1. Variation of settling velocity and residence with particle size

The settling velocity of small particles increases approximately quadratically with particle size, consistent with low Reynolds number particles with linear drag. As particle size increases to hundreds of microns and millimeters, the Reynolds number becomes larger, the drag becomes nonlinear, and the velocity therefore increases at a slower rate, eventually reaching a sublinear increase with particle size (figure 2). The particle settling velocities generally partition by density, with the dense PVC settling fastest. The standard deviations for the each polymer show the importance of shape. There is a large overlap in ribbons for PMMA and PET, even though PMMA has a specific gravity in freshwater of 1.187 and PET has a specific gravity of 1.396. Comparing the settling velocities of the particles with the characteristic velocities of relevant vertical transport processes in the oceans, it is clear that larger particles of PET, PMMA, and PVC cannot be kept in the water column by fronts, breaking waves, or convective cells alone, though stronger Langmuir circulation could counteract the settling velocity of these particles. Vertical motion induced by submesoscale fronts, on the other hand, may be able to keep PS particles smaller than 2 mm from settling but only PET particles smaller than around 500 μm. Because lakes are less energetic, the characteristic velocities of vertical processes are smaller and therefore one would expect the size of denser particles in the water column to be smaller. For example, average convective velocities found in lakes around the world (Read *et al* 2012) would only counteract the settling of PET particles less than 200–500 μm (figure 2(b)). There is a large range in the settling velocity of the PET particles with different shapes, whereas the PS particles have a much narrower range of settling velocities for a given size (figure 2).

The range of settling velocities spans four orders of magnitude for particles in the 1–5000 μm range (figure 2). The smallest particles (1–50 μm) sink on the order of 1 to 10 meters per day, whereas 5 mm particles can sink as fast as 10 000–15 000 meters per day. This large range in settling velocities leads to a large range in the expected residence time of different sized particles (figure 3). Very small particles could take over a year to sink 4000 m (approximately the average depth of the Pacific Ocean), while particles of PVC, PET and PMMA larger than 500 μm would take a day or less (figure 3(a)). PS particles, which are closer to neutrally buoyant, stay in the water longer, with 500 μm particles taking around a month to sink and 1500 μm particles settling in a day (figure 3(a)). If we consider a depth of 200 m, which is slightly less than the maximum depth of Lake Ontario (244 m), particles greater than around 40–60 μm of PVC, PET, and PMMA would be expected to sink within a month and particles greater than 100 μm would sink within a day (figure 3(b)). The PS particles again take longer to sink, with 100 μm particles taking around a month and particles greater than around 250 μm taking less than a day. (figure 3(b)).

This analysis further supports the idea that considering 'small' and 'large' MPs separately is appropriate (Sonke *et al* 2022), as the smallest MPs act more like tracers. By tracers, we mean the MP particles which have settling/rise velocities smaller than the characteristic vertical fluid velocities; these particles are also small enough to have low inertia relative to the flow (e.g. see Chamecki *et al* (2019), DiBenedetto *et al* (2023)). In this tracer limit, the settling/rise velocities are so low that the particles can become well-mixed within the water column. Note that this definition means there does not exist a universal cut-off for what constitutes a tracer MP, but that it depends on the specific flow environment.

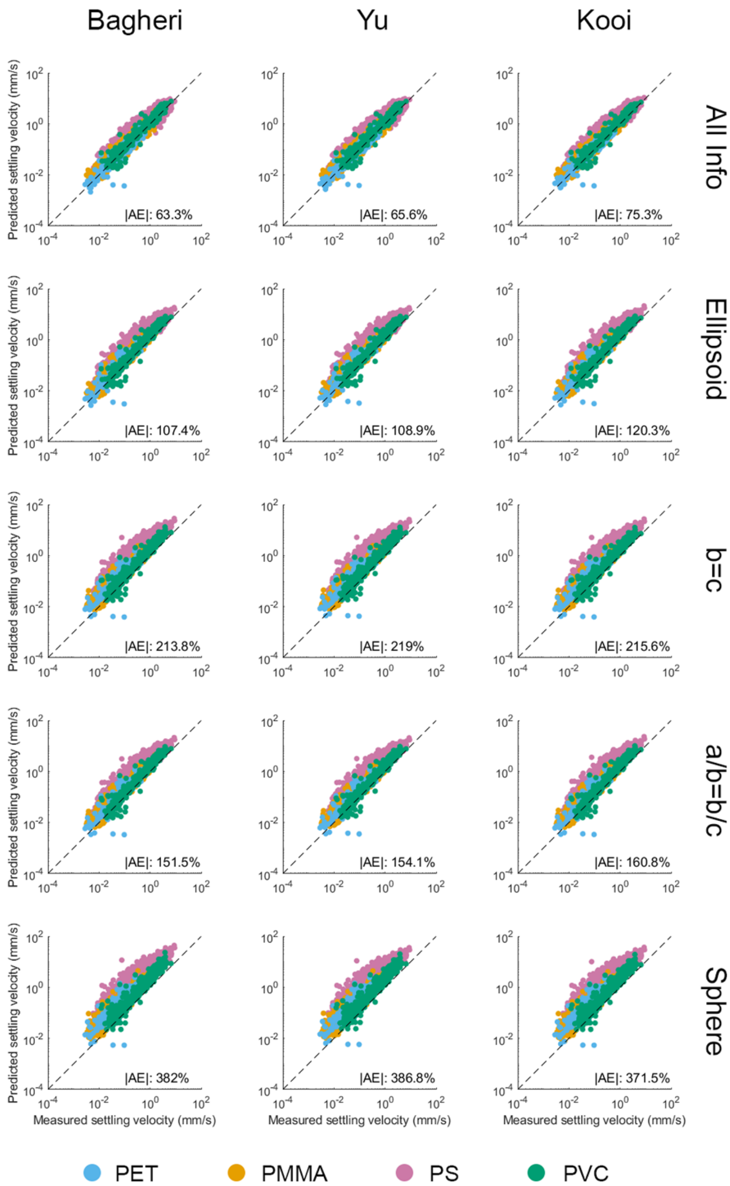

Figure 1. Comparisons of modeled particle settling velocity with measured settling velocity using three different equations and different assumptions about what particle parameters are known.

Particles larger than this size have large enough settling/rise velocities that they can easily settle out of the water column, or rise to the surface, without staying well-mixed.

3.1.2. Buoyant particles observed distribution case study

Here, we consider how the observed PSD can differ from the global PSD due to differential vertical transport. First, we consider a PSD sampled by a manta net at the free surface. We assume the measured depth of the net is a constant thickness δ. Given the equilibrium depth distribution of all of the particles in equation (6), we can calculate the fraction of particles of size d captured:

$$f(\mathrm{d}, \delta) = \int_{-\delta}^{0} g(z|\mathrm{d})\,\mathrm{d}z = 1 - \exp\left(-\frac{w_s(d)\,\delta}{K}\right). \tag{8}$$

The relative PSD observed from within the surface-layer is then

$$p_{\mathrm{obs}}(d, \delta) = p_0(d)\, f(d; \delta) = Cd^{-\alpha}\left[1 - \exp\left(-\frac{w_s(d)\,\delta}{K}\right)\right]. \tag{9}$$

This equation has a governing lengthscale ratio: δ/L_m where $L_m = K/w_s$ and represents the relative depth over which particles are mixed below the surface. In the limit $\delta/L_m \ll 1$ (i.e. sampling much

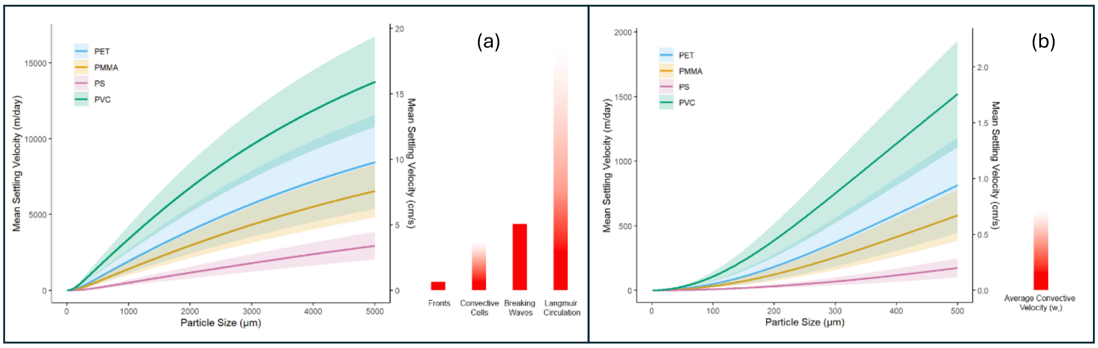

Figure 2. (a; left) The mean settling velocity, in m/day of the different polymers from 1–5000 μm along with the standard deviation as a function of particle size as compared to (right) the characteristic velocities of several important vertical transport mechanisms detailed in Van Sebille *et al* (2020). The gradient indicates the range of values, whereas the solid bar is when one value was given. (b; left) The same figure as in (a), but with a size range from 1–500 μm compared to (right) the range of average convective velocities ($w_\star$) given in Read *et al* (2012). The range is 2.01-7.43 mm s^{-1}, with most of the 40 lakes studied in the 2–5.5 mm s^{-1} range.

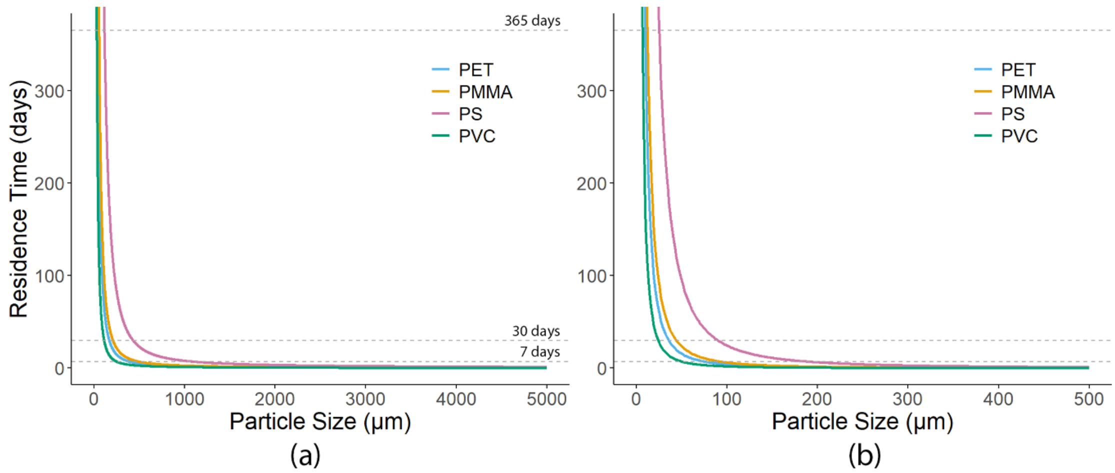

Figure 3. Average residence times for particles of different sizes assuming that they are settling (a) 4000 m, which is approximately the average depth of Pacific Ocean and (b) 200 m, which is approximately the maximum depth of Lake Ontario. Note the difference in the *x*-axis scale for both figures. The dashed lines represent, from bottom to top, a week, a month (30 days), and a year.

shallower than the depth over which the particles are mixed), $f \approx w_s(d_p)\delta/K$. In that case, $p_{obs}(d) \propto d^{-\alpha} w_s(d)$ (for constant K and δ). For small particles with rise velocities in the Stokes regime, $w_s \propto d^2$ (see Supplemental Information for exact equation). This implies an apparent scaling of

$$p_{obs}(d) \propto d^{-(\alpha-2)} \tag{10}$$

over the size range that is not fully sampled. However, for $\delta/L_m \gg 1$ (i.e. sampling much deeper than the depth over which the particles are mixed) one is able to sample the entire PSD and recover the original power law $p_{obs} \approx p_0$.

We apply this equation to an idealized scenario of spherical particles with constant specific gravity $\rho_p/\rho_f = 0.95$. Their rise velocities are calculated using the Stokes and Schiller-Naumann drag coefficients for $Re < 1$ and $Re > 1$ respectively (Clift *et al* 2005). We consider relatively mild turbulent diffusivity ($K = 10^{-5}$ m^2 s^{-1} and $K = 10^{-4}$ m^2 s^{-1}); note that K can approach 10^{-2} m^2/s and higher for near-surface MPs under wind and waves, but will generally decay over depth (Brunner *et al* 2015, Onink *et al* 2022). For the prescribed PSD, we use $\alpha = 2.5$ (which is similar to values in Kaandorp *et al* (2021), Kooi *et al* (2021)); however this same analysis can be applied to any α value. We plot the total PSD and the sampled PSDs for 4 different δ values (1, 5, 10, and 20 cm) in figure 4. In this figure, we also mark the particle size where $\delta = L_m$. Particles smaller than this critical size are sampled at the reduced power-law $\alpha - 2$, whereas particles larger than this are sampled at the true power law α. As sampling depth δ

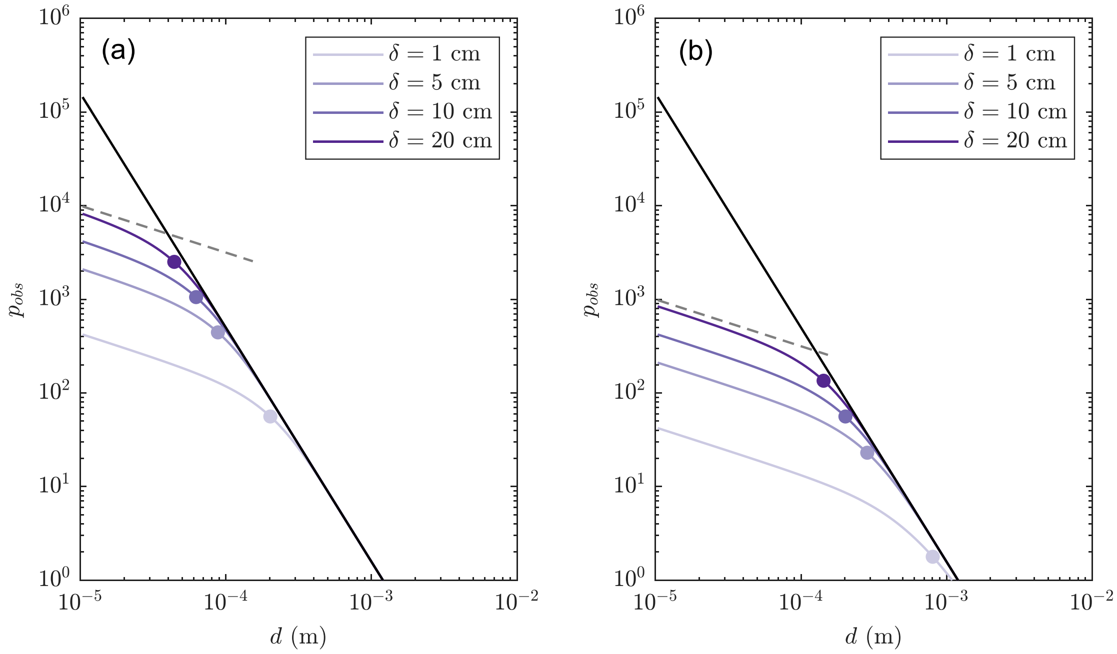

Figure 4. Observed PSD (in arbitrary units) of buoyant spheres as a function of surface sampling depth δ under constant turbulent mixing ((a) $K = 10^{-5}\mathrm{m}^2\,\mathrm{s}^{-1}$ and (b) $K = 10^{-4}\mathrm{m}^2\,\mathrm{s}^{-1}$). Solid line denotes $p_0 = d_\mathrm{p}^{-\alpha}$ and dashed line denotes $d_\mathrm{p}^{-\alpha+2}$. Dots mark particle size where the sampling depth is equivalent to the mixing length: $\delta = L_m$, which corresponds to the change in power law scaling.

increases, the critical size decreases and more of the original PSD is recovered. However, as diffusivity K increases, this critical size increases. In other words, under higher mixing, more of the water column needs to be sampled to recover the true PSD.

We show a similar analysis, but in this case consider nonspherical particles. We use the same scaled Dittmar *et al* (2024) fragment data to construct a synthetic PSD of particles that follows a power law for d_eq. We randomly choose particle geometry (a, b, c) from the Dittmar data and linearly scale it to the model distribution of d_eq. We choose particle specific gravity from a uniform random distribution of $\rho_\mathrm{p}/\rho_\mathrm{f}$ over the range 0.9 to 1. When modeling the rise velocities, we consider three different models: the Bagheri equation with known a, b, c; sphere with diameter given by d_eq; and sphere with diameter given by a. We plot this in figure 5. In this case, they are all sampled at the same depth $\delta =$ 20 cm under $K = 10^{-4}$ m^2 s^{-1}. We see that again, the smaller particles are sampled at the reduced power-law, whereas the larger particles recover the true power-law. There is some variability between the PSDs for the different assumed rise velocities, and there is a rightward shift in the PSD when using a rather than d_eq for the spherical diameter. However, the magnitude of the under-sampling due to mixing of smaller particles away from the free surface gives a much larger difference (multiple orders of magnitude for the smallest particles) than that of the different rise velocity assumptions.

3.1.3. Settling particles observed distribution case study

We next consider how the observed PSD changes as negatively buoyant particles settle out over time. We consider the same scaled Dittmar *et al* (2024) fragment data as in the previous section, but instead choose particle specific gravity from a random distribution between 1 and 1.1. We use the Bagheri and Bonadonna (2016) model to estimate settling velocity. We simulate particles which initially start uniformly distributed vertically over a water column that is 10 m deep; the particles then settle under constant turbulent mixing given by diffusivity $K = 10^{-4}$ m^2 s^{-1}.

We plot the PSD of the particles over time (1 h total) in two ways in figure 6: the PSD constructed from the particles still within the 10 m water column, and the PSD constructed from all particles which have settled below 10 m. This is designed to probe how observed PSDs vary as particles settle out of the water column, and compare to observed PSDs at depth/on the seafloor which are supplied by particles settling from above. First, for the PSD of particles still in the water column (figure 6(a)), we see that the PSD is approximately constant for the smallest particles, whereas the PSD steepens for the largest

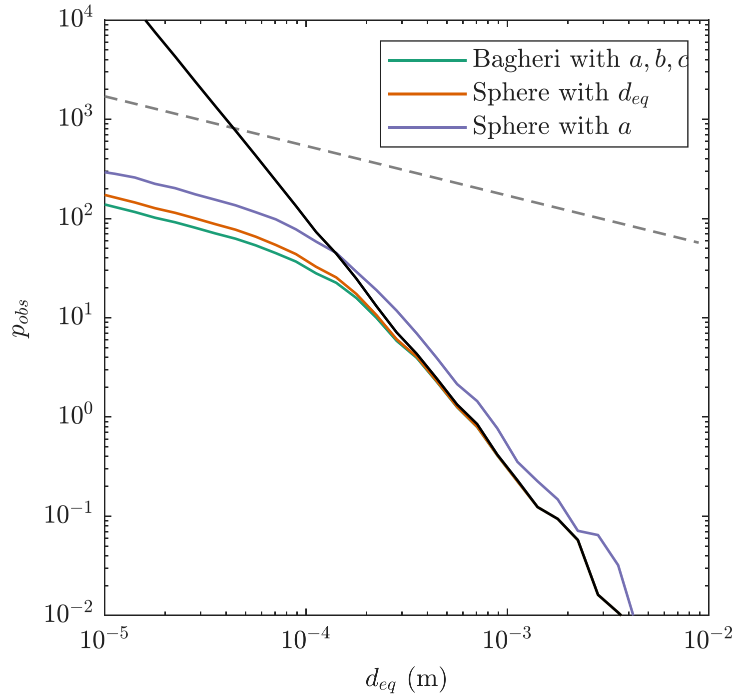

Figure 5. Observed PSD from simulated surface sampling with $\delta = 20$cm. Rise velocities of nonspherical particles computed using either Bagheri (Bagheri and Bonadonna 2016) model with full particle geometry, or sphere model assuming a diameter of either d_{eq} or a. Turbulent diffusivity is constant at $K = 10^{-4}$ m^2 s^{-1}. Solid line denotes $p_0 = d_{\mathrm{p}}^{-\alpha}$ and dashed line denotes $d_{\mathrm{p}}^{-\alpha+2}$.

particles. This difference arises because the smallest particles stay suspended and can be measured accurately, while the largest particles settle out and are under-estimated. This means that if only larger particles were sampled, the derived α value for the fragmentation power law would be overestimated.

In contrast, the PSDs measured from the particles which have settled out in figure 6(b) have three distinct regimes. The largest particles recover the true PSD because they have all settled out and can be measured accurately. The smallest particles follow the correct PSD slope but have a lower concentration. This is because they mainly diffuse out of the water column as their settling velocity is too slow; diffusion acts on all particles at the same rate, so the proper PSD slope is recovered. However, the intermediate-sized particles have only partially settled out, and therefore their PSD slope is controlled by their size-dependent settling velocity, as in the buoyant particle case study. Thus, we see a reduced power-law relative to the true PSD due to the differential vertical transport.

4. Discussion

4.1. Implications for settling velocity modeling

When comparing the performance of settling velocity models using different amounts of MP size and shape information, it is clear that measurements of three-dimensional particle shape are more important than the choice of settling velocity model used. This indicates that, when modeling the settling of real particles in aquatic systems, the current emphasis on developing additional high-accuracy models for MP settling velocity may be less impactful than improving measurements of MP geometry. Models that rely on increasingly detailed particle shape data will always be difficult to implement given that such data is unlikely to exist in typical environmental samples. Common MP samples in the literature at most include major and minor axis measurements, with many studies only reporting a single particle diameter. From the modeling side, it appears that, given accurate measurements of the particle size, shape, and density, both general fluid-mechanics based models (e.g. Bagheri and Bonadonna (2016)) and newer 'MP-specific' settling velocity estimates (e.g. Yu *et al* (2022)) perform similarly. Moreover, this study only considered uncertainty due to particle geometry, but it is worth noting that there is additional uncertainty due to particle density that can also affect the predicted settling velocity.

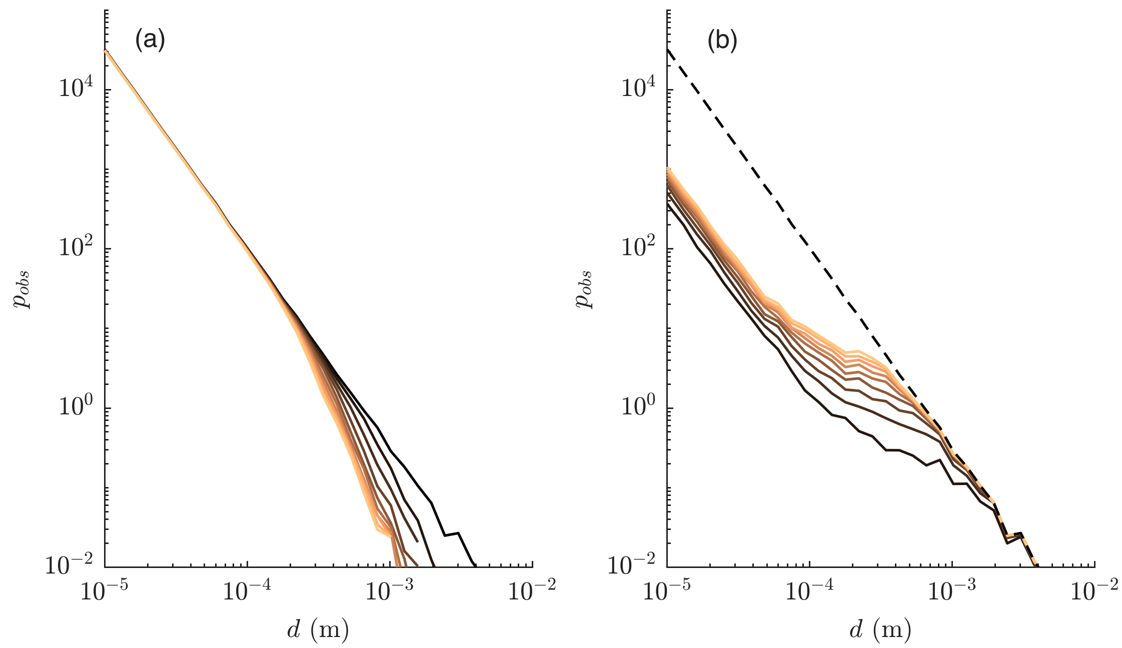

Figure 6. Observed PSD over time for simulated settling particles. The PSD is plotted over a time period of 1 h (dark to light). (a) Observed PSD integrated over the whole water column of 10 m. (b) Observed PSD of particles which have settled out of the water column. The dashed line shows p_0 for reference.

While there is still some room for improvement in model predictions—as seen by the fact that even with precise geometry information the best performing models have an absolute average error greater than 60%—if the goal is to model MP settling velocity to understand the transport and fate of MPs in the environment, then developing more accurate models of settling velocity using measurements that are available in the literature may be more beneficial. For the smallest MPs, improving shape data is likely less important, given that their settling and rise velocities are so small relative to characteristic fluid velocities that they behave like fluid tracers. Beyond particle geometry, there is also a need for models that take the age and transformation of the MPs into account.

4.2. Implications for environmental observations

From the observational side, the results of this study point to a need to record and publish as much information as possible on the particle geometry when measuring MPs in environmental samples. Presently, it appears that better data is more likely to improve the modeling of plastic transport in aquatic systems than improvements to the settling velocity equations. The good news is that these detailed measurements are becoming easier to collect. As more effort is placed on automated spectroscopy methods that utilize computer vision techniques to identify and size particles on a filter (Song *et al* 2021, Yang *et al* 2023), geometry information on MP samples is likely to be more readily available in the future. We encourage researchers collecting and analyzing environmental MP samples to publish high quality shape data.

4.3. Implications for PSDs and risk assessment

As discussed in the introduction, assumptions about the PSD of MPs in different environmental compartments are a crucial part of current risk assessment frameworks. This is because the smaller size fraction of MPs that is most biologically relevant is not well quantified by current analysis methods, meaning that the amount of small ($<50\,\mu$m) MPs must be estimated from counts of larger MPs which are easier to measure. Many current risk assessment and harmonization frameworks implicitly assume that MPs of different sizes are retained together within an environmental compartment, such that the local PSD is controlled only by fragmentation processes.

However, the orders of magnitude differences in the settling and rise velocities across MP size classes imply that the presence of larger MPs may not reliably predict the presence of small MP due to their distinct transport properties. In particular, the smallest MPs behave approximately as tracers and are therefore expected to be relatively uniformly mixed over the water column. Larger settling particles tend to accumulate at depth, while larger buoyant particles preferentially accumulate near the surface. As a result, samples collected near boundaries are expected to be enriched with larger particles, whereas samples collected within the water column should contain relatively more small particles. This conceptual picture is consistent with water-column observations reporting high concentrations of small MPs (Pabortsava and Lampitt 2020) and with surface observations that exhibit a roll-off in small-particle concentrations (e.g. as summarized in Kaandorp *et al* (2021)). It is also consistent with studies that show MPs segregating by rise velocity in the water column (DiBenedetto *et al* 2023) and segregation by size in sand (Dong *et al* 2018).

Our analytical and numerical results further suggest that size-dependent vertical transport introduces a characteristic cutoff in particle size, across which different apparent power-law exponents apply. For denser MPs, this cutoff arises from differential settling, which preferentially removes larger particles from the sampling volume while smaller particles remain suspended. As a result, power-law fits derived from observations dominated by larger particles can yield steeper exponents than those governing smaller size classes. This is consistent with observational studies where power-law fits based on measured size ranges predict higher abundances of small particles than are observed and are often attributed to measurement limitations (Kooi *et al* 2021). Our results indicate that this apparent discrepancy need not reflect measurement bias alone: preferential removal of larger non-buoyant particles can steepen the inferred power law, leading to systematic overestimation of small-particle abundances when extrapolated. This mechanism is evident in both the settling-particle simulations (figure 6(a)) and the buoyant surface-sampling experiments (figure 5). Under simplifying assumptions, the expected change in scaling can be derived analytically, with the apparent exponents differing by two across the transport-controlled size cutoff. More accurate measurements of small MPs are therefore needed to assess whether piecewise or size-dependent power-law descriptions better represent environmental PSDs.

4.4. General discussion

Taken together, the analyses presented in this study address how we model and interpret MP distributions; they suggest that rather than continuing to improve models for particle settling velocity, the community should focus instead on increasing the amount of available MP geometric data. In addition, our study indicates that the common practice of harmonizing environmental samples using a single power-law extrapolation is often physically inconsistent. Because the smallest MPs act as fluid tracers while larger particles preferentially settle or rise, the 'observed' size distribution at any given depth is not expected to match the underlying fragmentation spectrum. Our analysis assumed a constant turbulent diffusivity and neglected any biological or chemical interactions. Thus, while an idealized study, it isolates particle size and geometry as variables. Settling is only one of several processes that can produce size-dependent MP transport and abundance in the environment. For example, biological ingestion, aggregation, and biofouling may also contribute to size-selective sorting (Semcesen and Wells 2021). Although it is becoming increasingly common, we strongly recommend all studies consider separate size classes of MPs in both modeling and interpretation. Since the errors induced by reducing available information appear to be primarily a bias towards modeled particles sinking faster, there may be an opportunity to apply a bias correction to modeled sinking velocities to more accurately represent vertical transport. Finally, we encourage future work to test whether piecewise or size-dependent power-law descriptions better represent environmental PSDs, as such data would directly inform the validity of current harmonization and risk assessment approaches.

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Data availability statement

No new data were created or analysed in this study.

Supplementary information available at https://doi.org/10.1088/2515-7620/ae76df/data1.

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References

Bagheri G H, Bonadonna C, Manzella I and Vonlanthen P 2015 On the characterization of size and shape of irregular particles *Powder Technol.* **270** 141–53
Bagheri G and Bonadonna C 2016 On the drag of freely falling non-spherical particles *Powder Technol.* **301** 526–44
Borrelle S B *et al* 2020 Predicted growth in plastic waste exceeds efforts to mitigate plastic pollution *Science* **369** 1515–8
Brunner K, Kukulka T, Proskurowski G and Law K L 2015 Passive buoyant tracers in the ocean surface boundary layer: 2. observations and simulations of microplastic marine debris *J. Geophys. Res.: Oceans* **120** 7559–73
Chamecki M, Chor T, Yang D and Meneveau C 2019 Material transport in the ocean mixed layer: recent developments enabled by large eddy simulations *Rev. Geophys.* **57** 1338–71
Clift R, Grace J R and Weber M E 2005 *Bubbles, Drops and Particles* edn (Dover)
Cole M 2016 A novel method for preparing microplastic fibers *Sci. Rep.* **6** 34519
Coyle R, Service M, Witte U, Hardiman G and McKinley J 2023 Modeling microplastic transport in the marine environment: testing empirical models of particle terminal sinking velocity for irregularly shaped particles *ACS EST Water* **3** 984–95
Daily J and Hoffman M J 2020 Modeling the three-dimensional transport and distribution of multiple microplastic polymer types in Lake Erie *Mar. Pollut. Bull.* **154** 111024
Daily J, Tyler A C and Hoffman M J 2022 Modeling three-dimensional transport of and impacts of biofouling in Lake Erie and Lake Ontario *J. Gt. Lakes Res.* **48** 1180–90
DiBenedetto M H 2026 The fluid mechanics of ocean microplastics *Annu. Rev. Fluid Mech.* **58** 355–82
DiBenedetto M H, Donohue J, Tremblay K, Edson E and Lavender Law K 2023 Microplastics segregation by rise velocity at the ocean surface *Environ. Res. Lett.* **18** 024036
Dietrich W E 1982 Settling velocity of natural particles *Water Resour. Res.* **18** 1615–26
Dittmar S 2023 Additional data for 'settling velocities of small microplastic fragments and fibers (available at: https://doi.org/10.5281/zenodo.10049927)
Dittmar S, Ruhl A S, Altmann K and Jekel M 2024 Settling velocities of small microplastic fragments and fibers *Environ. Sci. Technol.* **58** 6359–69
Dong Z, Qiu Y, Zhang W, Yang Z and Wei L 2018 Size-dependent transport and retention of micron-sized plastic spheres in natural sand saturated with seawater *Water Res.* **143** 518–26
Fischer R, Lobelle D, Kooi M, Koelmans A, Onink V, Laufkötter C, Amaral-Zettler L, Yool A and van Sebille E 2022 Modelling submerged biofouled microplastics and their vertical trajectories *Biogeosciences* **19** 2211–34
Francalanci S, Paris E and Solari L 2021 On the prediction of settling velocity for plastic particles of different shapes *Environ. Pollut.* **290** 118068
Hataley E K, McIlwraith H K, Roy D and Rochman C M 2023 Towards a management strategy for microplastic pollution in the Laurentian Great Lakes-ecological risk assessment and management (part 2) *Can. J. Fish. Aquat. Sci.* **80** 1669–78
Isobe A, Kubo K, Tamura Y, Kako S, Nakashima E and Fujii N 2014 Selective transport of microplastics and mesoplastics by drifting in coastal waters *Mar. Pollut. Bull.* **89** 324–30
Jambeck J R, Geyer R, Wilcox C, Siegler T R, Perryman M, Andrady A, Narayan R and Law K L 2015 Plastic waste inputs from land into the ocean *Science* **347** 768–71
Kaandorp M L A, Dijkstra H A and van Sebille E 2020 Closing the Mediterranean marine floating plastic mass budget: inverse modeling of sources and sinks *Environ. Sci. Technol.* **54** 11980–9
Kaandorp M L A, Dijkstra H A and van Sebille E 2021 Modelling size distributions of marine plastics under the influence of continuous cascading fragmentation *Environ. Res. Lett.* **16** 054075
Kaandorp M L, Lobelle D, Kehl C, Dijkstra H A and van Sebille E 2023 Global mass of buoyant marine plastics dominated by large long-lived debris *Nat. Geosci.* **16** 689–94
Kaiser D, Estelmann A, Kowalski N, Glockzin M and Waniek J J 2019 Sinking velocity of sub-millimeter microplastic *Mar. Pollut. Bull.* **139** 214–20
Khatmullina L and Isachenko I 2017 Settling velocity of microplastic particles of regular shapes *Mar. Pollut. Bull.* **114** 871–80
Koelmans A A, Redondo-Hasselerharm P E, Mohamed Nor N H and Kooi M 2020 Solving the nonalignment of methods and approaches used in microplastic research to consistently characterize risk *Environ. Sci. Technol.* **54** 12307–15
Kooi M, Primpke S, Mintenig S M, Lorenz C, Gerdts G and Koelmans A A 2021 Characterizing the multidimensionality of microplastics across environmental compartments *Water Res.* **202** 117429
Kooi M, van Nes E H, Scheffer M and Koelmans A A 2017 Ups and downs in the ocean: effects of biofouling on vertical transport of microplastics *Environ. Sci. Technol.* **51** 7963–71
Kukulka T, Proskurowski G, Morét-Ferguson S, Meyer D W and Law K L 2012 The effect of wind mixing on the vertical distribution of buoyant plastic debris *Geophys. Res. Lett.* **39** L07601
Laxague N J M *et al* 2018 Observations of near-surface current shear help describe oceanic oil and plastic transport *Geophys. Res. Lett.* **45** 245–9
Lebreton L C M, van der Zwet J, Damsteeg J-W, Slat B, Andrady A and Reisser J 2017 River plastic emissions to the world's oceans *Nat. Commun.* **8** 15611
Lebreton L, Egger M and Slat B 2019 A global mass budget for positively buoyant macroplastic debris in the ocean *Sci. Rep.* **9** 12922
Lobelle D, Kooi M, Koelmans A A, Laufkötter C, Jongedijk C E, Kehl C and van Sebille E 2021 Global modeled sinking characteristics of biofouled microplastic *J. Geophys. Res.: Oceans* **126** e2020JC017098
Mehinto A C *et al* 2022 Risk-based management framework for microplastics in aquatic ecosystems *Microplastics Nanoplastics* **2** 17
Morales-Caselles C *et al* 2021 An inshore-offshore sorting system revealed from global classification of ocean litter *Nat. Sustain.* **4** 484–93

Onink V, Jongedijk C, Hoffman M, van Sebille E and Laufkötter C 2021 Global simulations of marine plastic transport show plastic trapping in coastal zones *Environ. Res. Lett.* **16** 064053

Onink V, van Sebille E and Laufkötter C 2022 Empirical Lagrangian parametrization for wind-driven mixing of buoyant particles at the ocean surface *Geosci. Model Dev.* **15** 1995–2012

Pabortsava K and Lampitt R S 2020 High concentrations of plastic hidden beneath the surface of the Atlantic Ocean *Nat. Commun.* **11** 4073

Read J S *et al* 2012 Lake-size dependency of wind shear and convection as controls on gas exchange *Geophys. Res. Lett.* **39** L09405

Rochman C M *et al* 2019 Rethinking microplastics as a diverse contaminant suite *Environ. Toxicol. Chem.* **38** 703–11

Semcesen P O and Wells M G 2021 Biofilm growth on buoyant microplastics leads to changes in settling rates: implications for microplastic retention in the Great Lakes *Mar. Pollut. Bull.* **170** 112573

Song Y K, Hong S H, Eo S and Shim W J 2021 A comparison of spectroscopic analysis methods for microplastics:manual, semi-automated and automated Fourier transform infrared and Raman techniques *Mar. Pollut. Bull.* **173** 113101

Sonke J E, Koenig A M, Yakovenko N, Hagelskjßr O, Margenat H, Hansson S V, De Vleeschouwer F, Magand O, Le Roux G and Thomas J L 2022 A mass budget and box model of global plastics cycling, degradation and dispersal in the land-ocean-atmosphere system *Microplast. Nanoplast.* **2** 28

Van Melkebeke M, Janssen C and De Meester S 2020 Characteristics and sinking of typical microplastics including the potential effect of biofouling: implications for remediation *Environ. Sci. Technol.* **54** 8668–80

Van Sebille E *et al* 2020 The physical oceanography of the transport of floating marine debris *Environ. Res. Lett.* **15** 023003

Van Sebille E, England M H and Froyland G 2012 Origin, dynamics and evolution of ocean garbage patches from observed surface drifters *Environ. Res. Lett.* **7** 044040

Van Sebille E, Wilcox C, Lebreton L, Maximenko N, Hardesty B D, Van Franeker J A, Eriksen M, Siegel D, Galgani F and Law K L 2015 A global inventory of small floating plastic debris *Environ. Res. Lett.* **10** 124006

Waldschläger K and Schüttrumpf H 2019 Effects of particle properties on the settling and rise velocities of microplastics in freshwater under laboratory conditions *Environ. Sci. Technol.* **53** 1958–66

Weiss L, Ludwig W, Heussner S, Canals M, Ghiglione J-F, Estournel C, Constant M and Kerhervé P 2021 The missing ocean plastic sink: Gone with the rivers *Science* **373** 107–11

Yang Z, Nagashima H and Arakawa H 2023 Development of automated microplastic identification workflow for Raman micro-imaging and evaluation of the uncertainties during micro-imaging *Mar. Pollut. Bull.* **193** 115200

Yu Z, Yang G and Zhang W 2022 A new model for the terminal settling velocity of microplastics *Mar. Pollut. Bull* **176** 113449